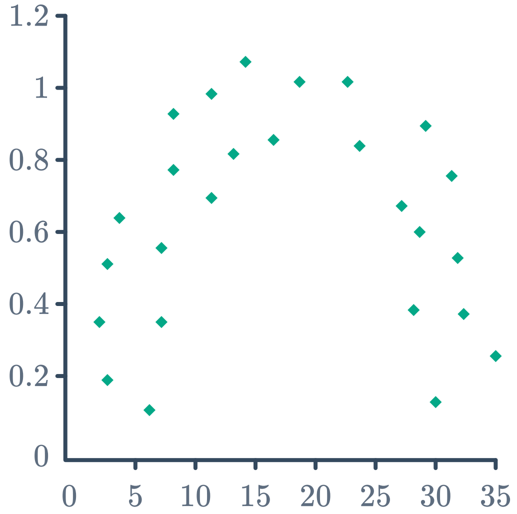
Correlation

1 For each of the following scatter plots, describe the data as linear or nonlinear:

a



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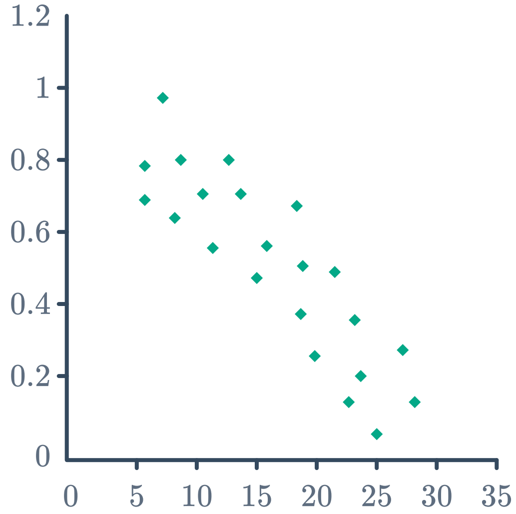

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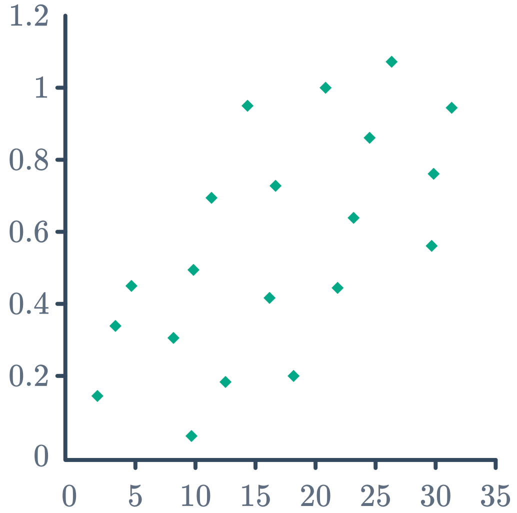

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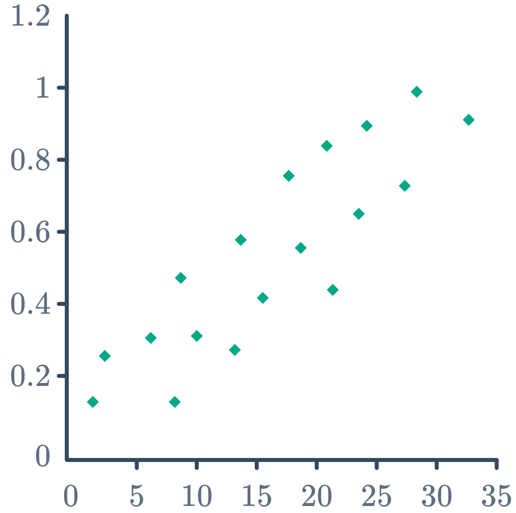

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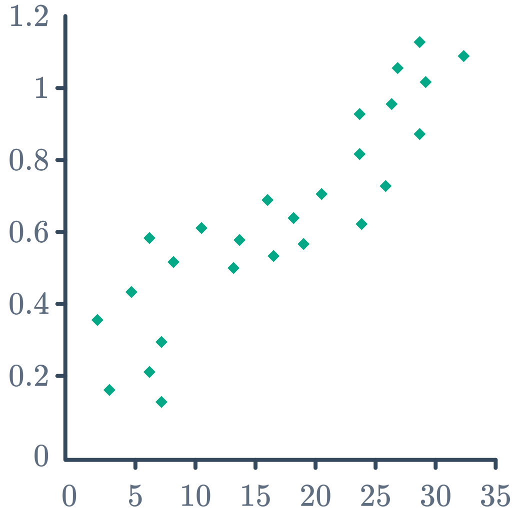

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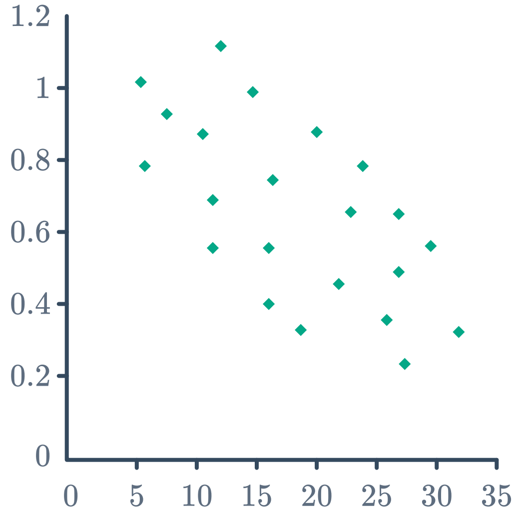
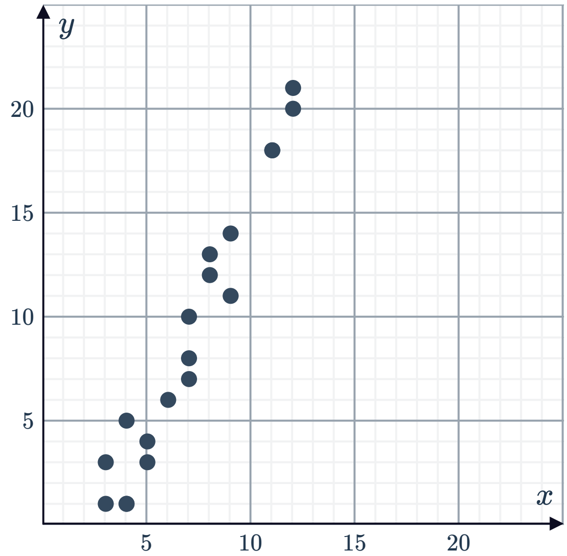


2 State whether the following scatter plots show a linear relationship, a nonlinear relationship, or no relationship:

a



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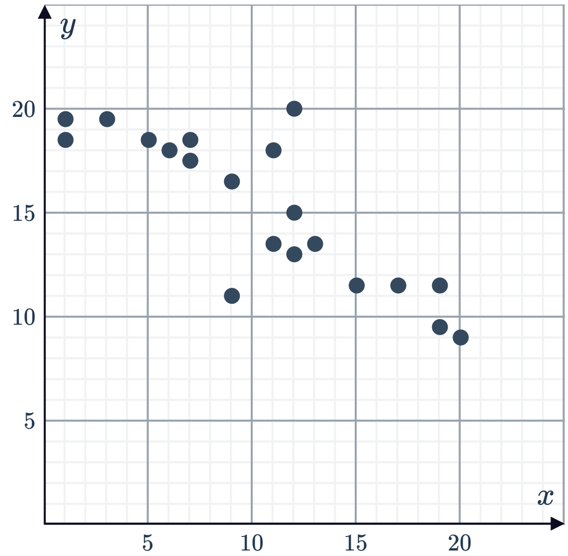

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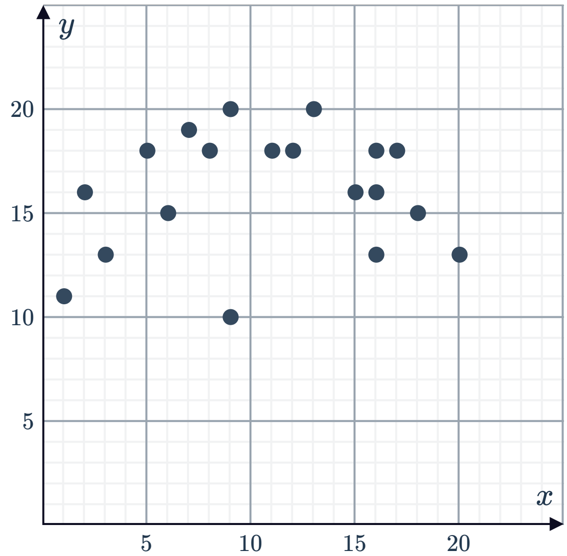

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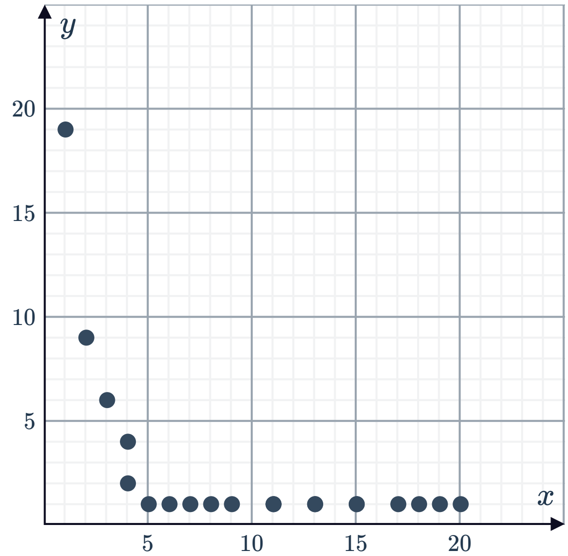

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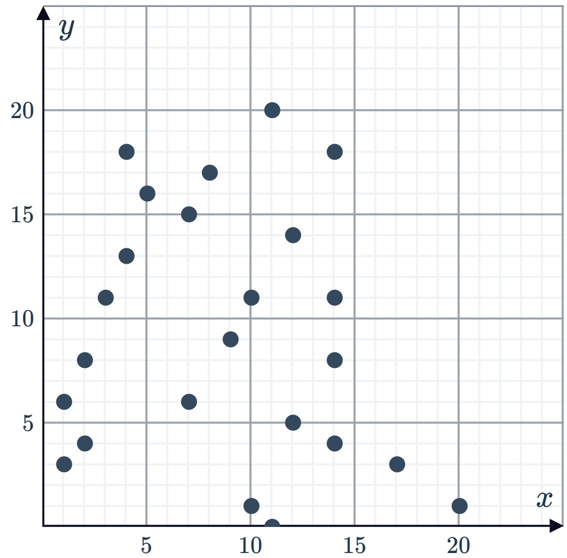

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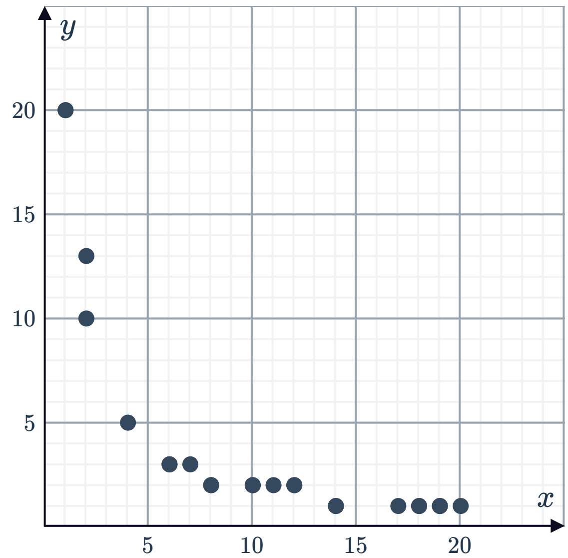

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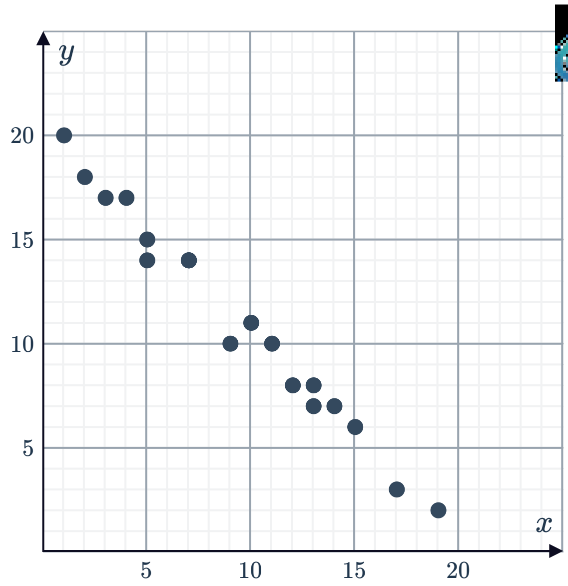
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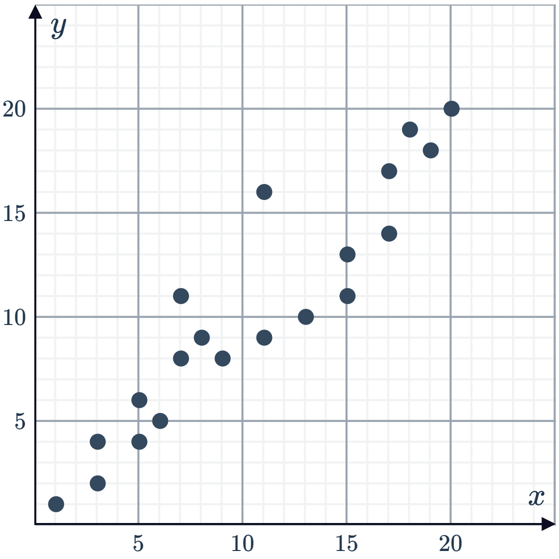
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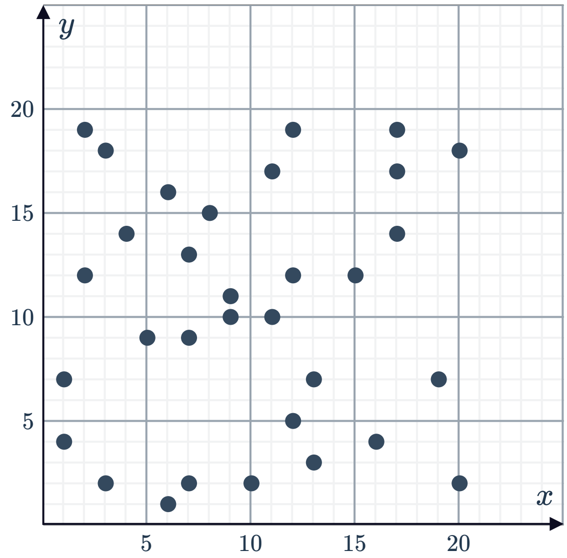
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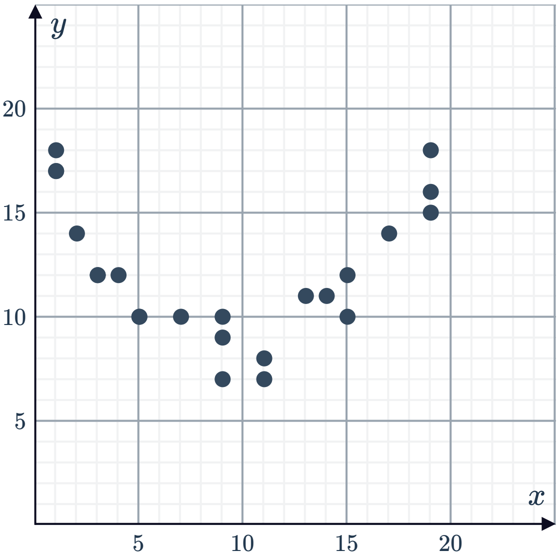
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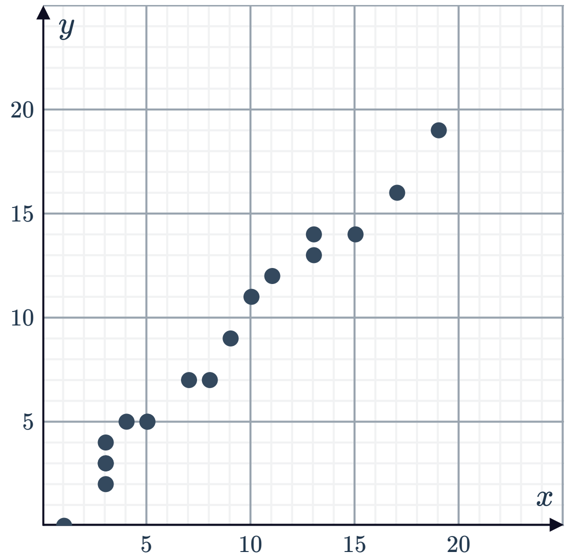
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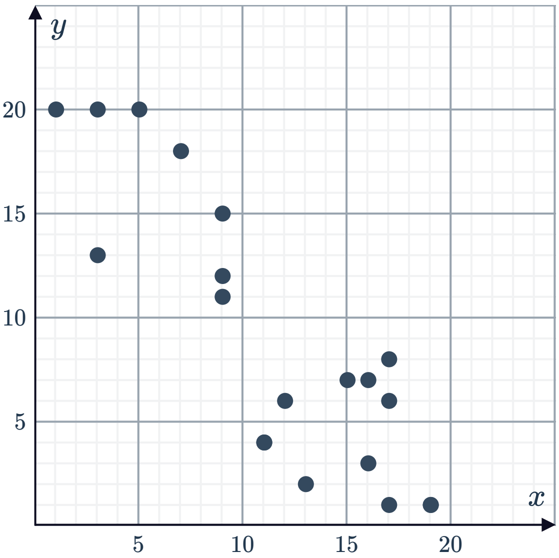
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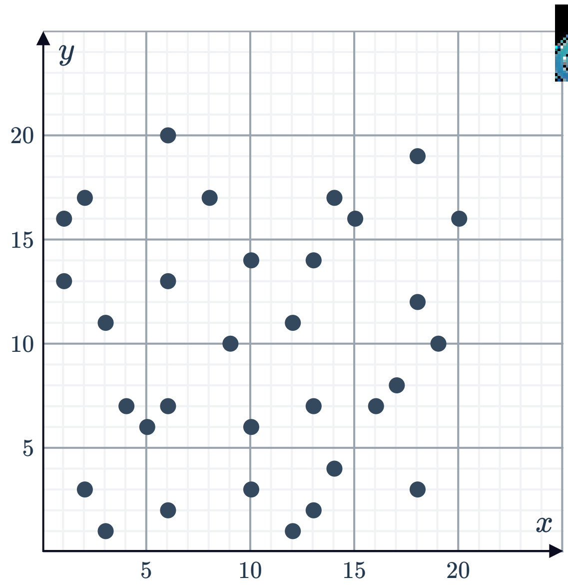

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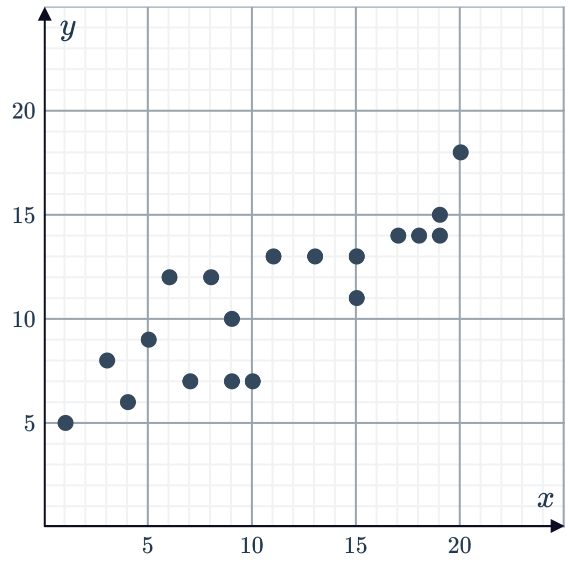
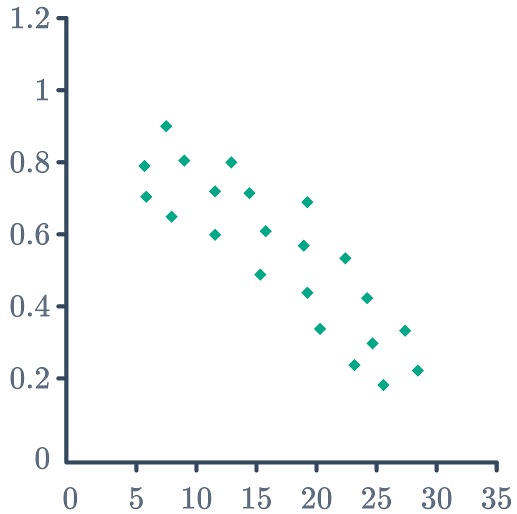
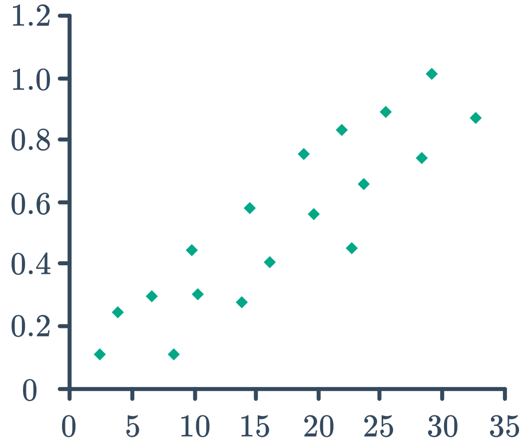


3 State whether the following scatter plots show a positive correlation, a negative correlation, or neither:

a



b



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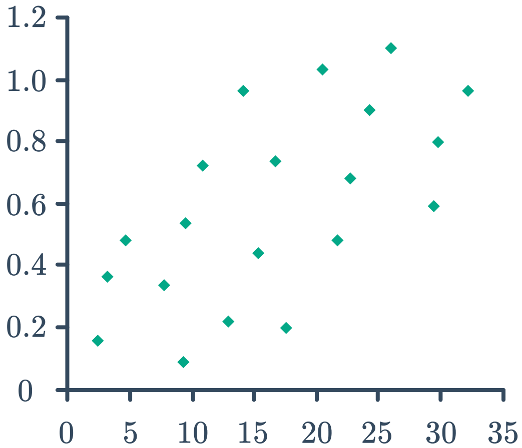

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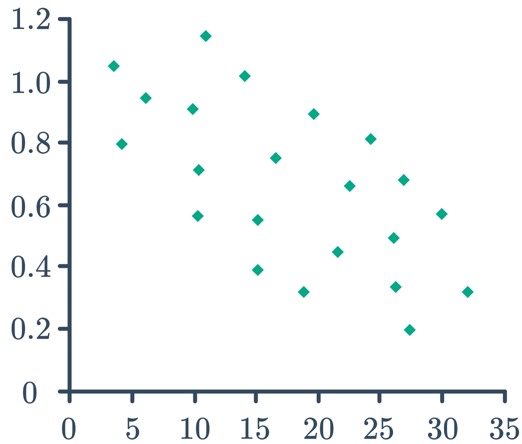

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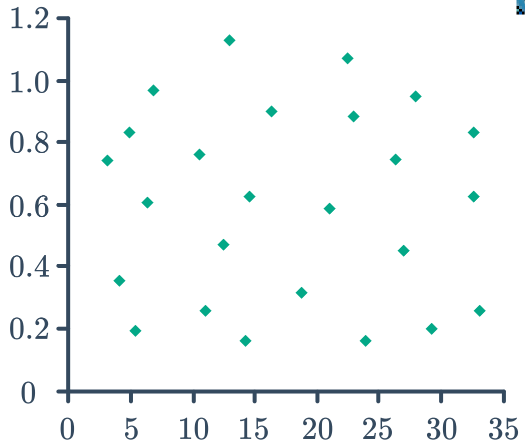
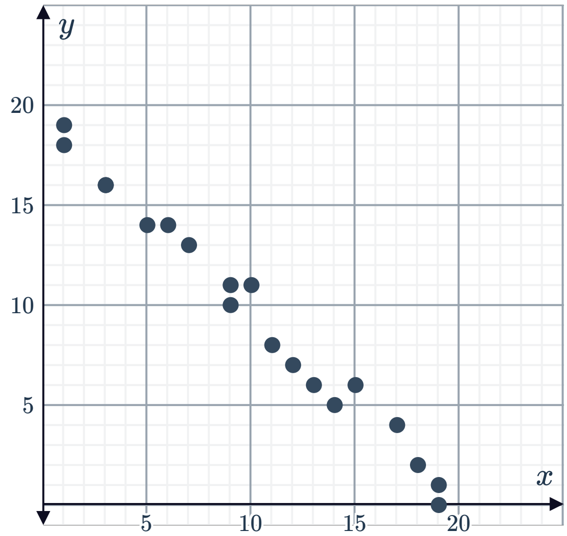
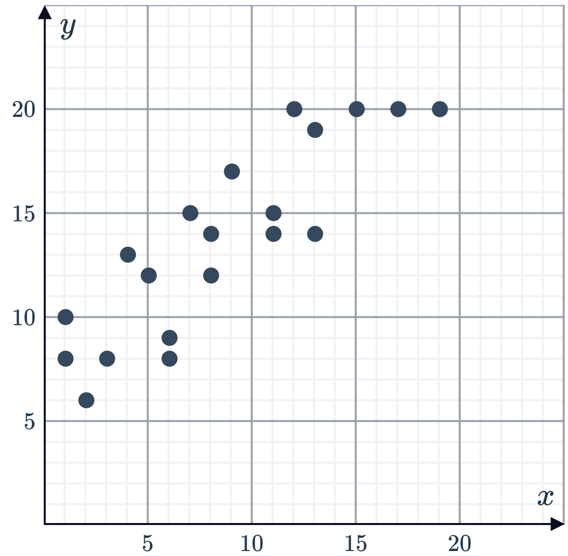


4 State whether the following scatter plots show a positive correlation, a negative correlation, or neither:

a



b



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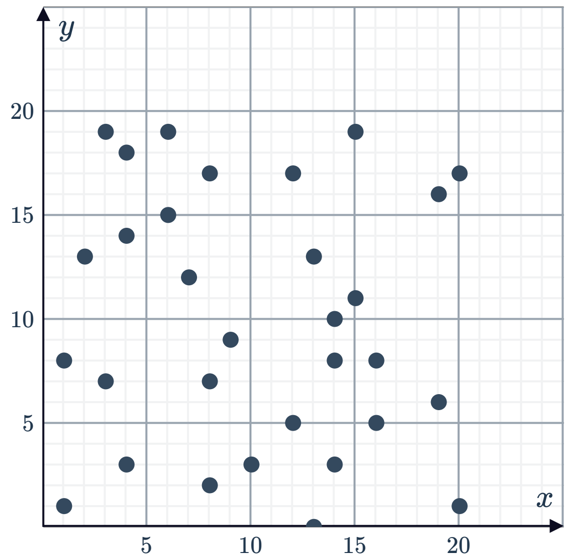

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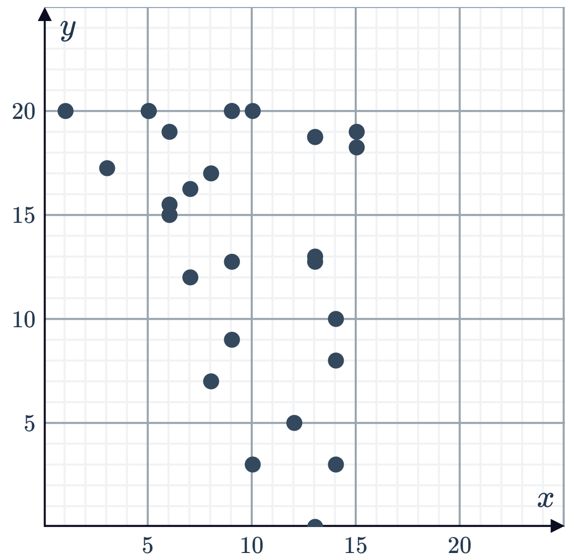

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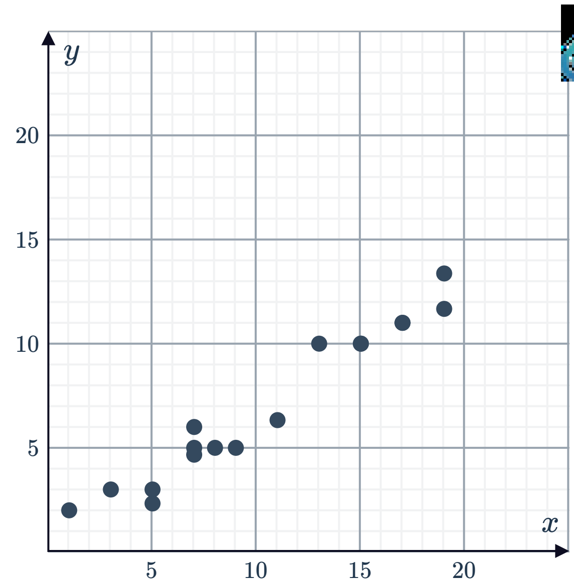
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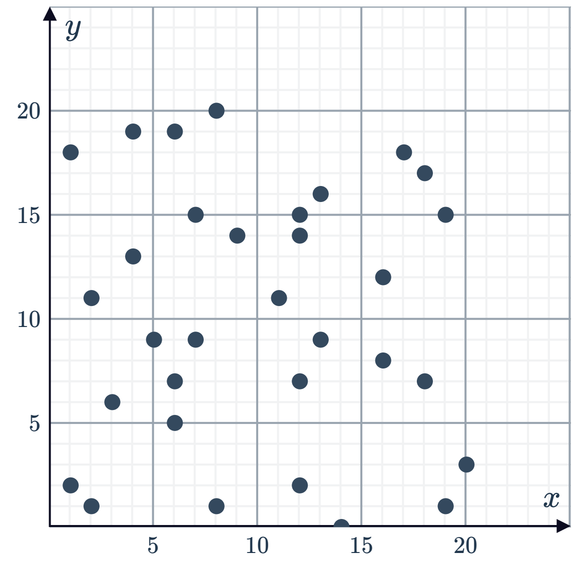
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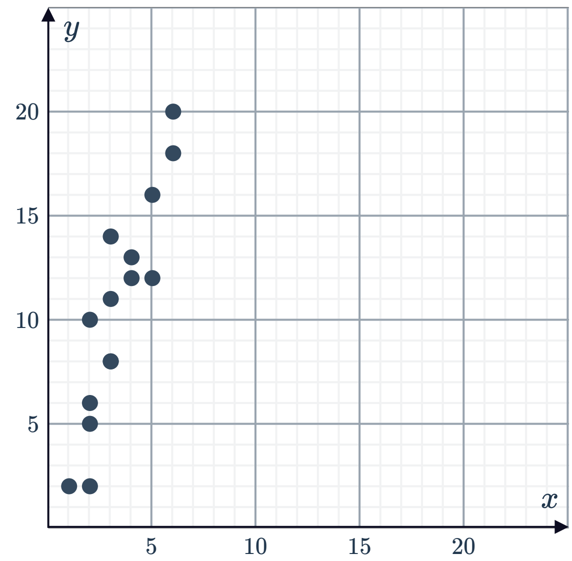
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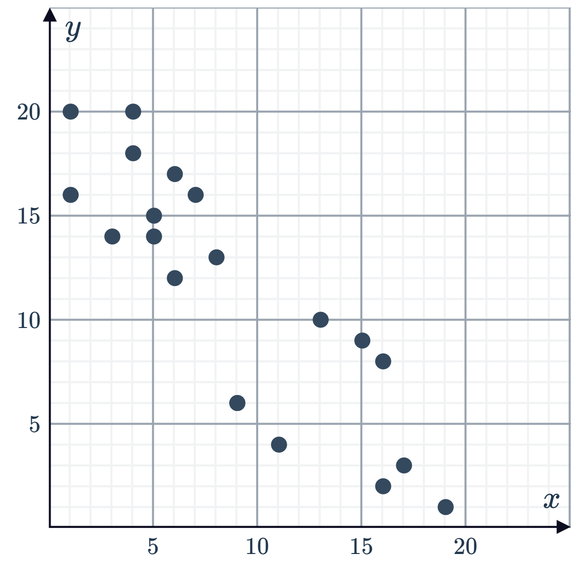
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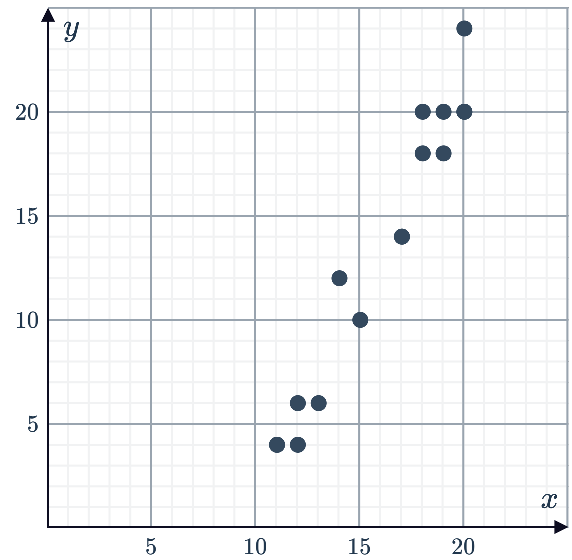
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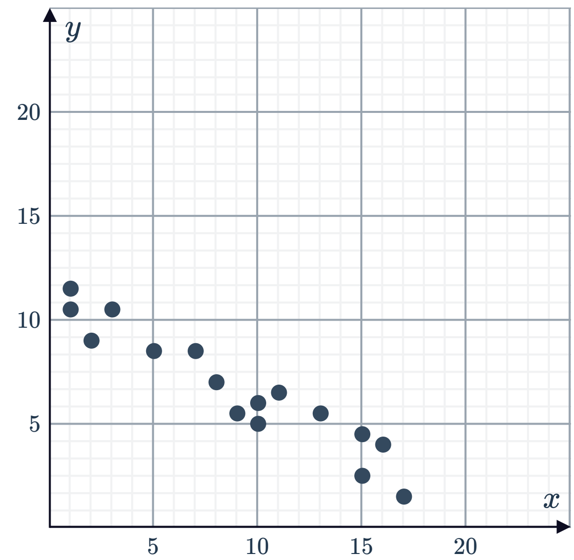
y
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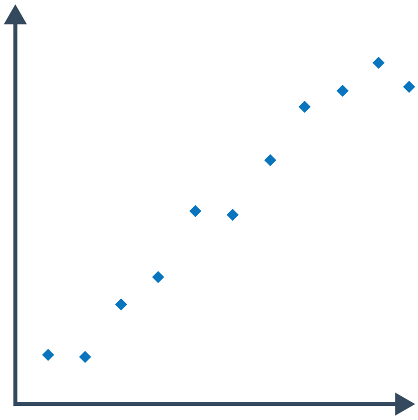
5 For each of the following scatter plots, describe the correlation in terms of strength and direction:

a



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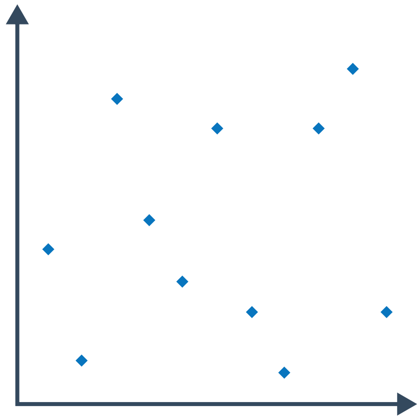

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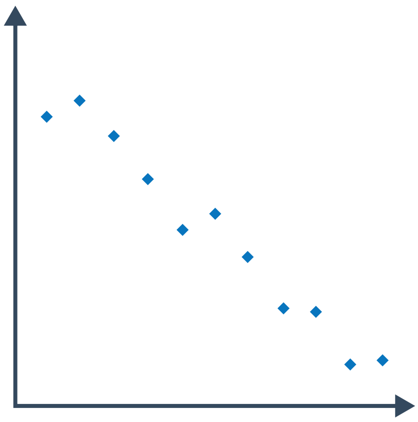

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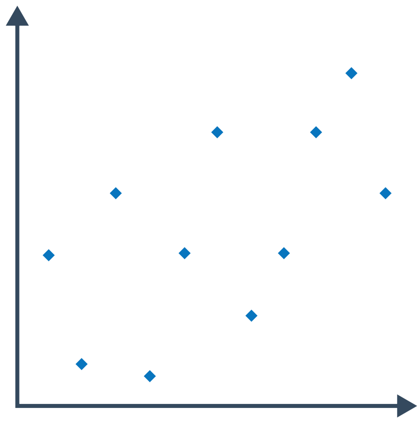

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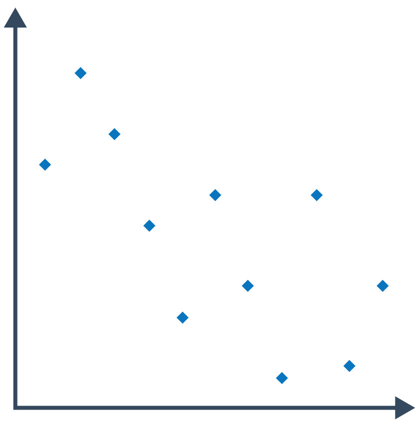
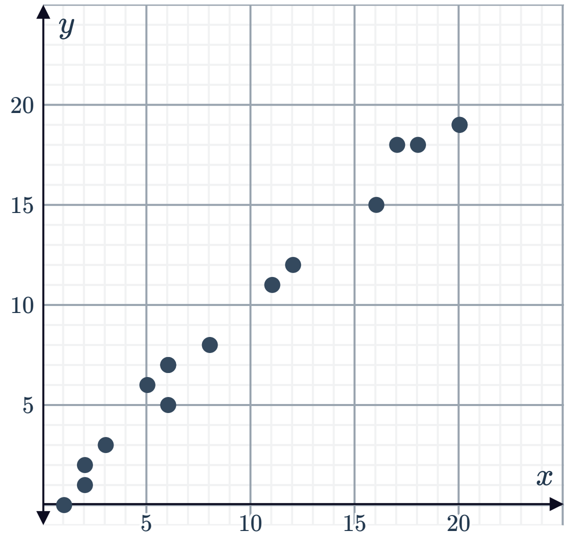


6 Describe the relationship between the var[redacted]s observed in the following scatter plots in terms of strength and direction:

a



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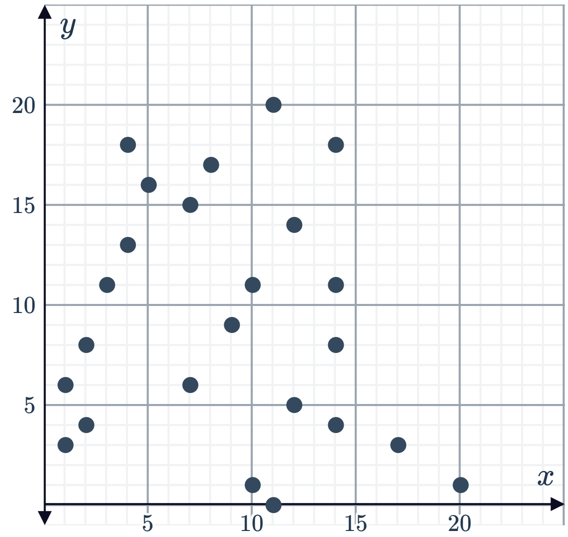

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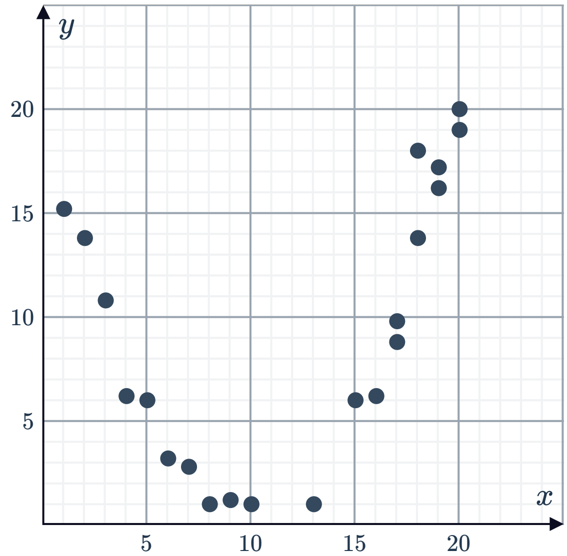

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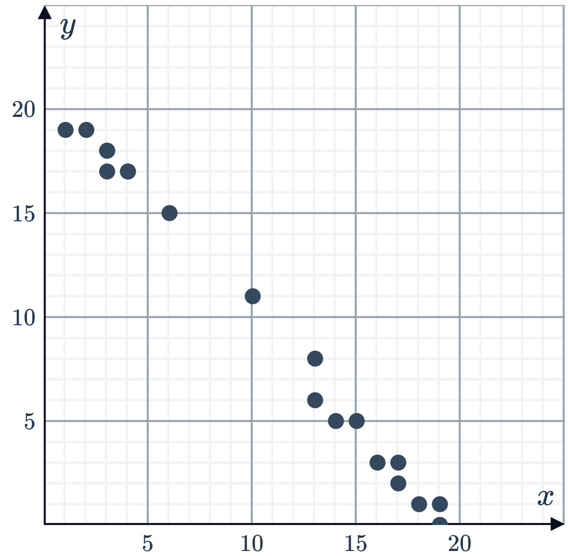

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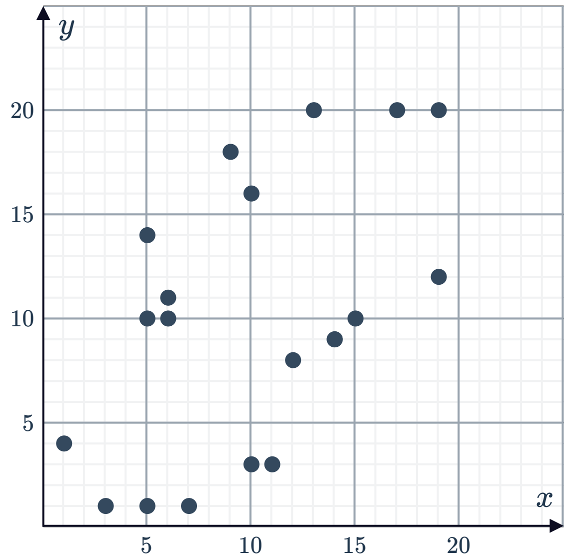

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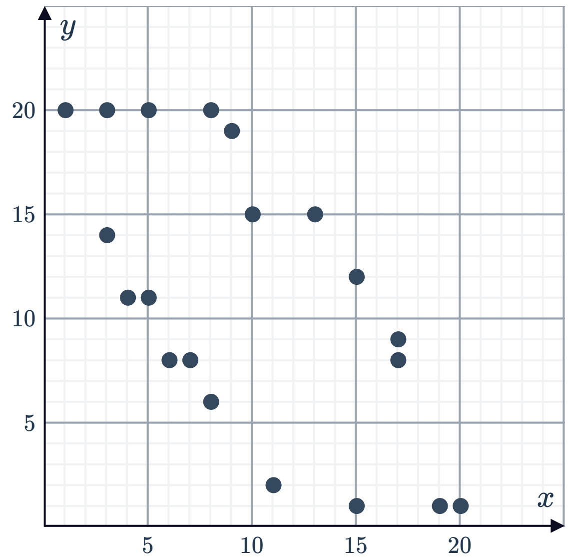


7 Describe the relationship observed between the variables in the following table of values in terms of strength and direction:

a

x	1	2	2	3	6	5	4	5	5	4
y	2	3	5	6	1	2	4	7	5	2

b

x	4	15	10	16	9	17	5	8	6	14
y	20	75	52	81	43	85	24	41	31	69

c

x	1	3	5	8	9	12	13	16	18	20
y	48	106	103	60	33	13	111	−6	38	66

d

x	46	25	35	95	39	96	17	45	24	86
y	4.2	2.6	3	9.7	4.4	9.5	1.5	4.7	2.7	8.6

e

x	11	8	12	14	17	20	9	13	3	15
y	−12	−7	−11	−14	−16	−21	−8	−13	−2	−14

8 For each of the following pairs of variables:

i Is there a relationship between the two variables. If yes, answer part (ii).

ii Is the correlation between the two variables positive or negative?

a Typing speed and dancing ability

b Temperature and ice cream sales

c Time spent studying and exam grade

d Eye colour and IQ

e Temperature and soup sales

f Rainfall and traffic accidents

g Height and weight

h Age and hearing ability

Applications

9 Determine whether the following are examples of variables with no correlation:

a The age of a child and their shoe size.

b The age of a child and their height.

c The age of a child and the number of pets owned.

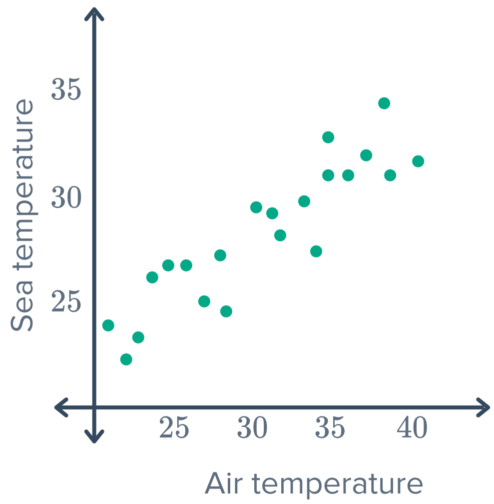
d The age of a child and the amount of adjectives learned.

10 Scientists conducted a study where each person was asked to read a paragraph and then recount as much information as they can remember. They found that the longer the paragraph, the less information each person could retain.
Is the correlation between the length of the paragraph and the information retained, positive or negative?

11 The scatter plot shows the relationship between air and sea temperature:

a Describe the relationship between air and sea temperature.

b Describe the correlation between air and sea temperature.



12 A database compare the income per capita of a country and the child mortality rate of that country. A sample of the data is shown in the table.
If a scatter plot was created from the entire database, describe the relationship you would expect it to have in terms of strength, direction and shape.

Income per capita	Child Mortality rate
1465	67
11 428	16
2621	35
32 468	9

13 A study was conducted to find the relationship between the age at which a child first speaks and their level of intelligence as teenagers. The following table shows the ages of some teenagers when they first spoke, and their results in an aptitude test:

Age when first spoke	14	27	9	9	16	21	17	10	7	19
Aptitude test results	96	69	84	90	101	87	92	99	104	93

a Construct a scatter plot for the data.

b Describe the correlation between the data points.

14 In recent years, beekeepers and scientists have become concerned over a phenomenon known as colony collapse disorder (CCD), where the majority of worker bees in a hive disappear, leaving behind the queen and immature bees. The percentage of beehive losses that can be attributed to CCD each year, since 2005, is shown in the table:

Year	Y	Hives lost to CCD, $(H)\%$
2005	0	18
2006	1	13
2007	2	23
2008	3	26
2009	4	28
2010	5	30

a Construct a scatter plot for this data.

b Describe the correlation between the number of years passed and the number of hives lost to CCD.

15 The market price of bananas varies throughout the year. Each month, a consumer group compared the average quantity of bananas supplied by each producer to the average market price (per unit).

a Construct a scatter plot using the data from the table.

b Describe the correlation between the supply quantity and the market price of bananas.

c According to this data, when will a supplier of bananas receive a higher price per banana?

Supply (kg)	Price (dollars)
550	15.25
600	14.75
650	14.75
700	14.75
750	14.25
800	14.00
850	13.75
900	13.25
950	13.50
1000	13.25

Outliers

16 The following table shows the number of traffic accidents associated with a sample of drivers in different age groups:

Age	20	25	30	35	40	45	50	55	60	65
Accidents	41	44	39	34	30	25	22	18	19	17

a Construct a scatter plot for this data.

b Describe the correlation between a person's age and the number of accidents.

c Is there an outlier in this data set?

17 The marks of 12 students in Maths and Sport were recorded in the following table:

a Construct a scatter plot for the students' marks in maths vs their marks in sports.

b Describe the correlation between students' marks in maths and marks in sport.

c Which student's scores appear to represent an outlier?

Student	Mark in Maths	Mark in Sports
1	63	44
2	92	74
3	60	52
4	79	70
5	88	67
6	81	60
7	61	73
8	91	86
9	72	84
10	42	93
11	66	57
12	92	92

18 A researcher is studying the relationship between the number of passers-by near an emergency situation and the time taken until help is offered. The results are shown in the table below:

a Construct a scatter plot to represent the data in the table.

b Comment on the relationship between the number of passers-by and the time until assistance is offered by a passer-by to a person in an emergency.

c Describe the correlation between the number of passers-by and the time until assistance is offered by a passer-by to a person in an emergency.

Passers-by (p)	Time until help is offered (t)
1	8
2	19
3	26
4	37
5	51
6	65

19 A study was conducted to compare running times in various outdoor temperatures. The table below lists the time taken to sprint 400 metres by runners in different temperatures:

Temperature (C)	5	2	10	8	1	7	6	4	3	9
Sprint time (s)	60	67	48	69	65	49	57	53	59	52

a Construct a scatter plot for this data.

b How many runners were tested in the study?

c Describe the correlation between temperature and sprint time for the data.

d Which data point represents an outlier?

20 The following table shows the average IQ of a random group of people against their height:

Height (cm)	140	145	150	155	160	165	170	175	180	185
IQ	103	95	98	111	85	89	108	145	110	93

a Construct a scatter plot for this data.

b Describe the relationship between IQ and height.

c How tall is the person who appears to be an outlier?